

3rd Feb 2026 / DAG360

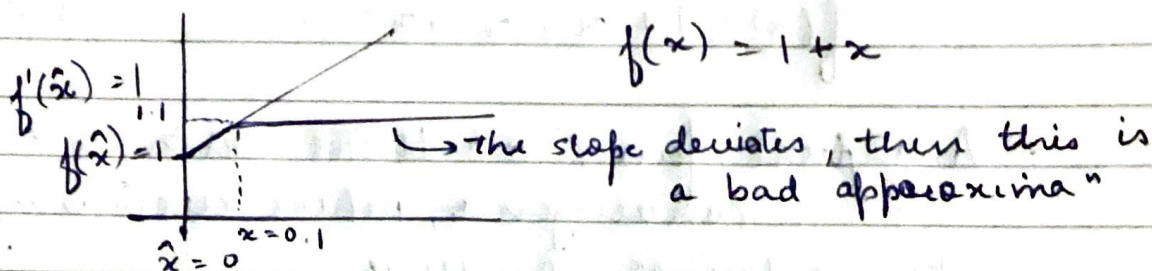
- why higher order is better?

it depends on $f(x) = f(\hat{x}) + f'(\hat{x})(x - \hat{x}) + \frac{f''(\hat{x})}{2}(x - \hat{x})^2$

$f'(\hat{x})$ eg. of funcⁿ whose 0th order is better than 1st order.

so, $\hat{x} = 0$

$$f(x) = f(\hat{x}) + f'(\hat{x})(x - \hat{x})$$



- small enough interval exists in which higher order approximation is better.

radius of convergence

e^x at $\hat{x} = 0 \Rightarrow$	$x = -2$	$0 < e^{-2} < 1$
0 th $\rightarrow 1$	1	
1 st $\rightarrow 1 + x$	-1	
2 nd $\rightarrow 1 + x + x^2/2$	?	

- Infinitely differentiable funcⁿ C^∞
 Analytical funcⁿ
 Entire Funcⁿ

n^{th} order

$$f(x) = f(\hat{x}) + f'(\hat{x})(x - \hat{x}) + \frac{f''(\hat{x})}{2}(x - \hat{x})^2 + \frac{f'''(\hat{x})}{6}(x - \hat{x})^3 + \dots + \frac{f^{(n)}(\hat{x})}{n!}(x - \hat{x})^n$$

eg. $\frac{1}{(1-x)}$ at $\hat{x} = 0$, $f^2(\hat{x}) = 1$

$$f'(\hat{x}) = \frac{1}{(1-x)^2} = 1$$

$$f''(\hat{x}) = \frac{2}{(1-x)^3} = 2$$

$$f'''(\hat{x}) = \frac{6}{(1-x)^4} = 6$$

$$f(x) = 1 + x + x^2 + x^3 + \dots \quad x \in [0, 1]$$

- Analytical funcⁿ $\rightarrow$ there exists a radius of convergence, where eventually enough terms things converge.
- Entire funcⁿ $\rightarrow$ entire x axis is the radius of convergence.

- Product Rule $\rightarrow$

$$\begin{aligned} f(x)g(x) &\approx (f(\hat{x}) + f'(\hat{x})(x-\hat{x})) \\ &\quad (g(\hat{x}) + g'(\hat{x})(x-\hat{x})) \\ &= f(\hat{x})g(\hat{x}) + f'(\hat{x})g(\hat{x})(x-\hat{x}) \\ &\quad + g'(\hat{x})f(\hat{x})(x-\hat{x}) \\ &\quad + f'(\hat{x})g'(\hat{x})(x-\hat{x})^2 \end{aligned}$$

- Chain Rule

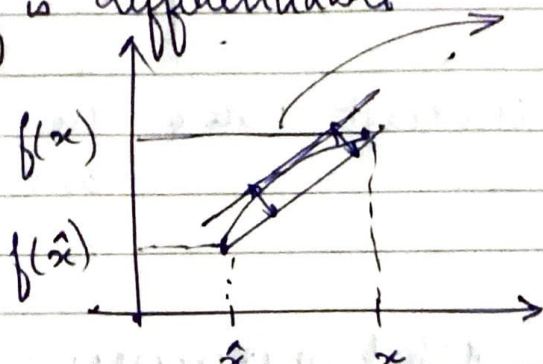
$$\begin{aligned} f(g(x)) &\approx f(g(\hat{x}) + \overset{\text{for } g}{g'(\hat{x})(x-\hat{x})}) \\ &= f(g(\hat{x})) + f'(g(\hat{x}))(g(x) - g(\hat{x})) \end{aligned}$$

$$\approx f(g(\hat{x})) + f'(g(\hat{x})) (g(\hat{x}) + g'(\hat{x})(x - \hat{x}) - g(\hat{x}))$$

→ Mean Value Theorem

f is differentiable

this proof comes from intermediate value theorem.



$\exists \bar{x} \in (\hat{x}, x)$ s.t.

$$\frac{f(x) - f(\hat{x})}{x - \hat{x}} = f'(\bar{x})$$

$$f(x) = f(\hat{x}) + f'(\hat{x})(x - \hat{x}) + \dots + \frac{f^{(n)}(\bar{x})(x - \hat{x})^n}{n!}$$

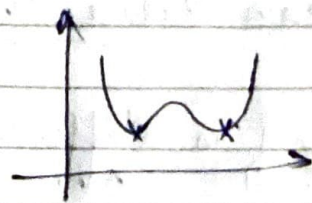
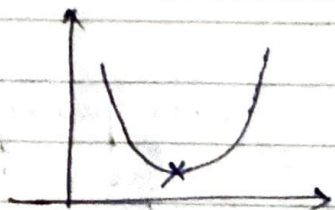
changes.

still there is no guarantee, you may add more more terms

- converg's base 13 funcⁿ → works concurrently to the intermediate value thm.

→ Characterisation of Minima

global minimum



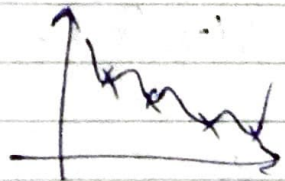
or



the least value the funcⁿ can take!

$x_* \in \mathbb{R}$ is global minimiser if $f(x_*) \leq f(x) \forall x$.

local minima



x_* is a local minimiser if $f(x_*) \leq f(x) \forall x \in (x_* - \delta, x_* + \delta)$
nearby

strict local minima

$$f(x_*) < f(x) \forall x \in (x_* - \delta, x_* + \delta)$$

→ First Order Necessary Condⁿ (FONC)

Assumpⁿ → f is diff., f' is continuous

If x_* is a local minimum, then $f'(x_*) = 0$

Proof → suppose not, i.e., $f'(x_*) \neq 0$
⇒ x_* is not a local min.

⇒ we can have a better pt.

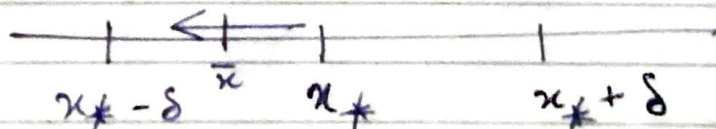
say $f'(x_*) > 0$ (0.1)

since $f'(x_*)$ is continuous, for $\epsilon = 0.01$, I can always choose a $\delta > 0$, s.t. $x \in (x_* - \delta, x_* + \delta)$

$$0.01 - 0.01 < f'(x) < 0.01 + 0.01 \quad (\text{values are illustrative})$$

∃ a $\delta > 0$, s.t. $\underbrace{f'(x)}_{\text{remains positive}} > 0 \quad \forall x \in (x_* - \delta, x_* + \delta)$

$$f(x) = f(x_*) + \underbrace{f'(\bar{x})}_{+ve} \underbrace{(x - x_*)}_{-ve}$$



$$x \in (x_* - \delta, x_*)$$

$$f(x) < f(x_*)$$